5	(a)	(i)	No <u>net</u> heat flow/transfer between the objects. [B1] Both objects have the same temperature.[B1]
		(ii)	The thermodynamic temperature scale: • has an absolute zero, and [B1] • is independent of the property of any particular substance. [B1]
	(b)	(i)	At low pressures, the gas <u>particles/molecules/atoms</u> are far apart from each other. [B1] Hence, the forces of attraction between the gas particles (or intermolecular forces) can be considered to be <u>negligible</u>. [[B1]
		(ii)	$U = \frac{3}{2}NkT$ For one mole, $U = \frac{3}{2}N_A kT$ $= \frac{3}{2}\left(6.02\times10^{23}\right)\left(1.38\times10^{-23}\right)\left(120+273.15\right)$ [C1] $= 4900$ J [A1]

6	(a)	(i)	Emission line spectrum consists of discrete bright coloured lines on a